Multiple Disease Prediction in Crops using Deep Learning

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Abstract—**A deep convolutional neural network was developed to predict multiple diseases in crop leaves by leveraging a large-scale labelled image dataset. The model demonstrated high performance across accuracy, precision, recall, and F1-score on the test set. These results indicate the efficacy of the proposed method for automated disease detection in agricultural settings, offering potential for early intervention and yield preservation.**

I. Introduction

Agricultural productivity is a cornerstone of global food security, with crop health playing a critical role in determining yield outcomes. Diseases in crops—arising from pathogens such as fungi, bacteria, and viruses—are among the most significant threats to sustainable agriculture, often causing up to 40% loss in global crop production annually. These losses are particularly detrimental in regions dependent on agriculture for economic and nutritional sustenance.

Traditionally, disease identification relies on manual inspection by trained agronomists or farmers, which is both labor-intensive and subject to high inter-observer variability. Such methods do not scale to large agricultural plots and often result in delayed diagnoses, reducing the window for effective treatment. Furthermore, visual symptoms of different diseases often exhibit significant overlap, making diagnosis non-trivial even for experts.

In this context, automated, data-driven techniques offer promising alternatives. With the advent of deep learning, particularly convolutional neural networks (CNNs), computer vision has witnessed significant advances in image-based classification tasks. Leveraging large-scale labelled image datasets, deep learning can be harnessed to develop models capable of robust, scalable, and real-time disease detection across various crop species. This potential is especially relevant for developing intelligent farming systems, which require high-throughput, accurate diagnostics to support precision agriculture. The present work investigates a CNN-based approach for predicting multiple crop diseases from leaf images, addressing the need for scalable and reliable plant disease monitoring tools.

II. Methodology

A. Dataset and preprocessing

The *PlantVillage* dataset [1] was utilized, comprising 54,303 labelled leaf images across 38 crop species and multiple disease classes. Each image was resized to 224×224 pixels, normalized to zero mean and unit variance, and subjected to online augmentation including random rotations, flips, and brightness adjustments to enhance generalization. A stratified train-validation-test split of 70:15:15 was applied to maintain class balance throughout training.

B. Model architecture

A stacked convolutional neural network architecture was constructed, featuring five convolutional blocks. Each block contained a convolutional layer (with a 3×3 kernel size and a padding of 1) followed by batch normalization and a ReLU activation followed by max pooling for spatial downsampling. The final convolutional output was flattened and passed through two dense layers before a softmax classifier produced probabilities for each disease category. Batch normalization and a dropout of 0.5 were incorporated to stabilize training and prevent overfitting.

C. Training setup

The model was trained using the Adam optimizer with an initial learning rate of 1×10^{-4} and categorical cross-entropy loss. Training proceeded for 10 epochs with a batch size of 32. Early stopping monitored validation loss with a patience of five epochs. Learning rate reduction by a factor of 0.5 was triggered upon plateauing validation accuracy.

III. Results and Discussion

A. Performance

On the held-out test set, the proposed stacked convolutional neural network achieved an overall accuracy of 98.28%. The accuracy and loss curves over during training indicate that the model plateued at around 8 epochs (Fig. 1).

The overall metrics are summarized in Table I. Class-wise precision averaged 98.30%, indicating that only 1.70% of all positive disease predictions were false positives. Recall reached 98.28%, demonstrating that the model missed just 1.72% of true disease instances. The harmonic mean of precision and recall yielded an F1-score of 98.27%, reflecting the model's strong balance between sensitivity and specificity.

TABLE I
OVERALL METRICS ON THE TEST SET

Metric	Score
Test Accuracy	0.9828
Test Precision	0.9830
Test Recall	0.9828
Test F1-Score	0.9827

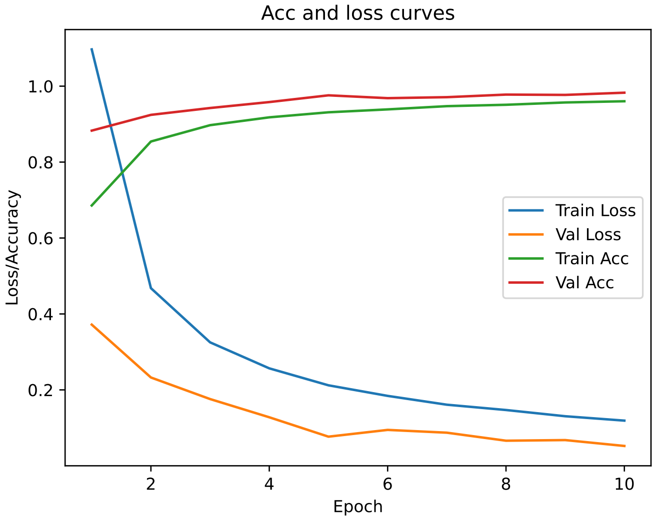

Fig. 1. Accuracy and loss curves

B. Discussion

The exceptionally high recall is particularly valuable for agricultural disease monitoring, as it minimizes false negatives and ensures that nearly all diseased plants are flagged for intervention. The slightly lower precision suggests occasional misclassification, most likely arising from symptom overlap among certain diseases that exhibit visually similar lesion patterns. In real-world deployments, this trade-off may necessitate a secondary verification step (e.g., expert review) to filter out false alarms.

IV. Conclusions and Future Work

The proposed stacked CNN model, trained on a curated and augmented dataset, demonstrates substantial capability in distinguishing complex visual patterns across disease classes, highlighting its potential as a valuable tool in precision agriculture.

Nevertheless, several avenues remain for further improvement. Future work will investigate the application of transfer learning from large-scale natural image corpora, which could enhance feature representations in scenarios with limited agricultural training data. Moreover, the integration of hyperspectral and multispectral imaging data may improve detection sensitivity to early-stage or latent infections that are not visually apparent in RGB imagery.

In addition, embedding the trained models in edge-computing platforms would enable real-time inference in field conditions, overcoming connectivity and latency issues prevalent in rural settings. The adoption of attention-based mechanisms and model ensembles is also anticipated to refine classification accuracy, particularly in cases involving overlapping symptomatology or inter-class ambiguity. Finally, active learning strategies could be used to iteratively expand the training dataset with expert feedback, ensuring model adaptability to evolving plant disease phenotypes and new crop varieties.

V. Related Work

The application of machine learning to crop disease detection has evolved rapidly with the advent of deep learning. Early systems, such as the Neocognitron and LeNet-5, demonstrated the feasibility of hierarchical feature extraction for pattern recognition in constrained domains like digit classification and simple leaf images. With the introduction of AlexNet [2], the field of image classification experienced a paradigm shift, setting the stage for deeper and more expressive convolutional architectures.

Chen *et al.* [3] employed transfer learning with pre-trained networks such as VGGNet and Inception for the identification of plant leaf diseases. This approach significantly reduced training time while achieving competitive accuracy. Ramcharan *et al.* [4] extended this idea to real-time diagnosis using mobile devices for cassava disease classification, highlighting the feasibility of field deployment.

Recent research has emphasized lightweight architectures tailored for low-power devices, such as SimpleNet [5], which optimized wheat disease detection without compromising performance. Comparative evaluations have consistently shown that CNN-based classifiers outperform traditional methods like support vector machines (SVMs), particularly in complex cases like rice sheath blight detection [6].

Despite these advancements, most existing models are constrained to single-label classification and limited disease taxonomies. The current study contributes by addressing multi-disease prediction using a custom deep architecture optimized for high-resolution plant imagery, marking a step toward more comprehensive and field-ready plant pathology solutions.

References

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